

Name: B. Ganga Jayadev Reddy

Reg. No.: 192472248

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1. What is Data Science? Explain its primary objective.

A) Data Science is the study of collecting, processing, and analyzing data to discover useful information. It uses statistics, programming, and machine learning to solve real-world problems.

Primary objective: To convert raw data into meaningful insights that help people and organizations make better decisions.

2. Why is Data Science important in today's digital world? Mention any four reasons.

- A)** • Supports better decision-making.
- Predicts future trends and customer behavior.
 - Improves business efficiency.
 - Provides personalized services to users.

3. List any four benefits of Data Science in business and society.

- A)** • Better business decisions.
- Early fraud detection.
 - Improved healthcare services.
 - Increased productivity and efficiency.

4. Mention four real-world applications of Data Science.

- A)** • Healthcare
- Online shopping recommendations
 - Banking fraud detection
 - Weather forecasting

5. What are the different facets of data? Briefly explain each.

- A)** • Structured Data – Organized in rows and columns.
- Semi-structured Data – Partially organized, such as XML and JSON.
 - Unstructured Data – Includes images, videos, and text.
 - Metadata – Information that describes other data.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

A) Structured: Student database.

Semi-structured: JSON or XML files.

Unstructured: Images, videos, emails.

7. What is Big Data? State its key characteristics (5 Vs).

A) Big Data refers to very large and complex datasets that cannot be handled using traditional tools.

The 5 Vs are Volume, Velocity, Variety, Veracity, and Value.

8. Explain the Big Data ecosystem. Name any four technologies used in it.

A) The Big Data ecosystem includes tools used to collect, store, process, and analyze data.

Examples: Hadoop, Spark, Hive, and HBase.

9. What are the major phases of the Data Science process?

A) Data Collection, Data Preparation, Data Analysis, Model Building, Model Evaluation, Deployment, and Presentation of Results.

10. What is data retrieval? Mention two common sources from which data can be retrieved.

A) Data retrieval is the process of obtaining data from storage systems.

Common sources are databases and APIs.

11. Why is data collection considered an important step in the Data Science process?

A) It ensures that the data used for analysis is accurate, relevant, and reliable, leading to better results

12. What is data cleansing? List any four common data-cleaning techniques.

A) Data cleansing is the process of correcting or removing incorrect data.

Techniques: Removing duplicates, handling missing values, correcting errors, and standardizing formats.

13. What is data integration? Why is it necessary in Data Science?

A) Data integration combines data from different sources into one dataset. It improves data quality and makes analysis more accurate.

14. Explain data transformation with suitable examples.

A) Data transformation means converting data into a suitable format for analysis.

Examples include converting text into numbers, changing date formats, and normalizing values.

15. What is data analysis? Mention any four techniques used for data analysis.

A) Data analysis is the process of examining data to identify useful patterns and information.

Techniques: Statistical analysis, data mining, machine learning, and data visualization.

16. Differentiate between descriptive, predictive, and prescriptive analytics.

A) Descriptive analytics explains what happened.

Predictive analytics estimates what may happen.

Prescriptive analytics recommends the best action to take.

17. What is meant by building a model in Data Science? Mention the basic steps involved.

A) Model building means creating a machine learning model using data.

Steps: Select data, choose an algorithm, train the model, test it, and improve its performance.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

A) Model evaluation checks how well a model performs before it is deployed.

Metrics: Accuracy, Precision, Recall, and F1-Score.

19. Why is presenting findings an essential part of the Data Science process?

Mention any four visualization tools or techniques.

A) It helps others understand the results clearly and supports better decision-making.

Tools: Bar chart, pie chart, line graph, and Tableau.

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

A) Models are deployed by integrating them into software, websites, mobile apps, or cloud platforms.

Examples: Netflix recommendations, credit card fraud detection, medical diagnosis, and traffic prediction.